Characterizing Spatial Variation in Internet Access Latency: A Multilevel Approach

Jonatas Marques*
IPinfo
Seattle, USA
jonatas@ipinfo.io

Jared N. Schachner
University of Southern California
Los Angeles, USA
jschachn@usc.edu

Nicole P. Marwell
University of Chicago
Chicago, USA
nmarwell@uchicago.edu

Nick Feamster
University of Chicago
Chicago, USA
feamster@uchicago.edu

Abstract

Crowdsourced datasets are vital for analyzing variation in Internet access performance across geographic areas and addressing these spatial disparities. Prior research using these data and pursuing similar objectives often examines performance variation between single spatial units, such as census tracts, and scrutinizes a limited set of sociodemographic variables like race or class composition to explain it. However, this approach may be insufficiently precise to characterize the patterns of, and explanations for, spatial differences in Internet performance. We argue that multilevel, multivariate models better represent Internet performance as a spatial phenomenon; these models decompose its variance at multiple spatial scales and permit inclusion of multiple explanatory factors at each level. To demonstrate the utility of this approach, we use multilevel, multivariate models to analyze spatial patterns of Internet latency (idle and under load) and jitter drawn from crowdsourced Ookla Speedtest data collected between 2022 and 2023. Despite prior research's emphasis on neighborhood variation in Internet performance, our multilevel models on crowdsourced data reveal that latency and jitter varies far more within neighborhoods than between them. Moreover, demographic differences in residential populations explain only a small portion of the modest neighborhood-level variance in Internet performance we estimate. Variation in infrastructure across counties and states appears to stratify performance to a far greater extent.

CCS Concepts

• **Social and professional topics** → **Broadband access**; • **Networks** → **Network measurement**; **Public Internet**.

Keywords

Broadband Access, Multilevel Modeling, Broadband Access Performance, Latency

*Work done while at the University of Chicago.



This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.
IMC '26, Karlsruhe, Germany
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2327-8/26/10
https://doi.org/10.1145/3777912.3809140

ACM Reference Format:
Jonatas Marques, Jared N. Schachner, Nicole P. Marwell, and Nick Feamster. 2026. Characterizing Spatial Variation in Internet Access Latency: A Multilevel Approach. In *Proceedings of the 2026 ACM Internet Measurement Conference (IMC '26), October 12–16, 2026, Karlsruhe, Germany*. ACM, New York, NY, USA, 14 pages. https://doi.org/10.1145/3777912.3809140

1 Introduction

Internet access has become a necessity of modern life. Those without reliable access to Internet may miss out on a range of crucial opportunities, including education, communication, commerce, social interactions, and healthcare. Although much prior research has explored the patterns and predictors of Internet access overall, recent work confirms that even among households that have access, the reliability and performance of their Internet connection varies widely. These differences may have material implications for users' ability to take full advantage of the Internet's benefits. Thus, precisely measuring levels of, and understanding variation in, Internet access networks' performance is essential [1, 7, 12, 31, 42]. This line of inquiry is particularly critical in a policy environment marked by substantial investments devoted to expanding access to reliable, high-speed broadband.

Crowdsourced data has become indispensable to this line of inquiry, allowing users to probe temporal and spatial variation in Internet reliability and performance. However, we argue that most prior research [7, 12, 13, 17, 23, 26, 31, 32, 39–41] using these sources has been limited in several ways. First, spatial variation in crowdsourced-derived Internet performance measures is typically analyzed based on only a single spatial unit (typically census tracts or zip codes to proxy neighborhoods). Yet there may be much more variation in performance within than between neighborhoods, rendering differences in neighborhood-level averages uninformative. Moreover, higher-order spatial structures (e.g., cities and counties) may stratify Internet performance to a greater extent than do neighborhoods. Regardless of the level at which prior spatial differences in Internet performance are measured, researchers typically explain these differences as a function of sociodemographic measures. But other variables (e.g., infrastructure and natural barrier proximity) may provide as much if not more explanatory power regarding spatial variation in Internet performance.

Clarifying the spatial scales at which Internet performance may vary most sharply, and identifying what variables at these spatial

scales are most strongly associated with high or low performance, lays crucial groundwork for improving our understanding of spatial disparities in Internet performance. For example, if there is more variation in Internet performance within urban neighborhoods than between them, or at higher-order spatial scales (e.g., state or county) than at the neighborhood level, such findings give rise to distinct considerations for designing sampling frames that will more precisely illuminate spatial disparities in Internet performance. Indeed, given the biases in crowdsourced performance data, how to design and deploy alternative sampling strategies to achieve these aims is a critically important task, albeit one that remains contested.

At the same time, our effort to clarify the spatial scales at which Internet performance may vary has important policy implications. For example, if performance differences between neighborhoods are small, while within-neighborhood performance variation is substantial, this would suggest that household- rather than neighborhood-level differences may need to be targeted by policy interventions. Such interventions might include programs that help individual households upgrade their network equipment (e.g., WiFi routers) and personal devices (e.g., smartphones, laptops). Similarly, if county-level variation in performance is substantial, then expanding middle-mile infrastructure within a state may be warranted as the most effective intervention.

To the best of our knowledge, this paper is the first to demonstrate the utility of specifying multilevel models on crowdsourced data to precisely characterize and explain spatial variation in Internet performance. To do so, we fit multilevel models to crowdsourced Internet performance data drawn from Ookla users across core urban areas in the United States in 2022 and 2023. We focus on latency (both idle and under load) and jitter as the outcome variables, as opposed to other conventional network metrics, such as throughput. Our focus reflects shifts in both Internet infrastructure and the factors of network performance that ultimately affect application performance and user experience. First, over the past decade, residential broadband speeds have increased substantially, with many households now accessing connections that far exceed the throughput requirements of common applications, such as video streaming. Concurrently, the applications that dominate modern Internet usage have become increasingly latency-dependent. Video conferencing, interactive gaming, and streaming services all rely critically on low latency to deliver acceptable user experience [49]. Even conventional "legacy" throughput-oriented applications such as web browsing have proven more sensitive to latency than to available bandwidth, with prior work demonstrating that latency reductions improve user engagement and outcomes more effectively than throughput increases [9, 44, 47]. Given these trends, latency has emerged as the primary constraint on Internet performance and the most salient metric for characterizing disparities in both application performance and, ultimately, user experience. We focus our work on two questions:

- RQ1: What proportion of the variance in Internet performance as represented by crowdsourced speedtests resides between (versus within) neighborhoods, cities, counties, and states?
- RQ2: What proportion of the measured neighborhood-level variance that exists is explained by neighborhood sociodemographic composition, versus other factors?

2 Background

This section describes multilevel modeling and its application to our work. Multilevel models (also called hierarchical linear models or mixed-effects models) analyze data with hierarchical or clustered structures [18, 24, 36, 43]. Hierarchical data appears in many fields and arises from sampling design. For example, in educational research, students may be nested within classrooms, which are themselves nested within schools. Similarly, in medical research, patients may be nested within physicians, within departments, and ultimately within hospitals. Multilevel models capture this layered structure and determine how much each level of the hierarcy contributes to outcome variance [43].

We use multilevel analysis to understand how and why Internet performance varies across spatial units. Multilevel models assume hierarchical data, with one outcome measured at the lowest level (e.g., individual) and explanatory variables at all levels (e.g., individual and group) [18]. Crowdsourced datasets like Ookla Speedtest have nested structures well-suited to this approach. Speedtest results nest within devices, connections, buildings, neighborhoods, cities, and so forth. A simple linear model does not capture this hierarchical structure. Multilevel models partition variance between hierarchy levels, revealing the relative importance of each level. Each level has potential explanatory variables: device capacity, connection utilization, building access technology, neighborhood plan availability and affordability, etc. Multilevel analysis estimates individual-level effects, group-level effects, and cross-level interactions simultaneously. This approach reveals contextual effects, specifically how group characteristics influence individual outcomes—essential to understanding Internet performance differences and informing interventions that bridge the digital divide.

Multilevel models thus extend linear models by incorporating fixed population-level effects and random group-specific variations. Consider students nested in schools. A two-level model examines how student background (e.g., socioeconomic status, study hours) and school characteristics (e.g., teacher-student ratio, funding) affect student outcomes (e.g., math scores) [35]. Within each school j, a regression equation describes outcome Y for student i from student background variable X and random error. The *within-school* model appears in Equation 1. Here, Y_{ij} is the outcome for student i in school j, X_{ij} is a student characteristic, β_{0j} and β_{1j} are regression parameters, and r_{ij} is random error (normally distributed with zero mean).

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + r_{ij} \tag{1}$$

Two parameters summarize each school's outcome distribution: intercept β_{0j} and slope β_{1j}. If X is centered on the school mean, β_{0j} is the mean outcome for that school. β_{1j} represents the effect of student background on outcome. Small β_{1j} indicates weak association between X and outcome; large β_{1j} indicates strong differentiation by student background.

These parameters characterize each school's outcome distribution. Multilevel analysis studies how these parameters vary across schools. The *between-school* model (Equation 2) serves this purpose.

$$\beta_{0j} = \gamma_{00} + \gamma_{01} W_j + V_{0j} \tag{2a}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11} W_j + V_{1j} \tag{2b}$$

Here, W_j is a school characteristic, γ_{00}, γ_{01}, γ_{10}, and γ_{11} are regression parameters, V_{0j} is the unique increment to average outcome for school j, and V_{1j} is the unique increment to slope. Both V_{0j} and V_{1j} are normally distributed with zero means and covariance.

Combining Equations 1 and 2 yields the mixed model (Equation 3).

$$Y_{ij} = \gamma_{00} + \gamma_{01} W_j + \gamma_{10} X_{ij} + \gamma_{11} W_j X_{ij} + (V_{0j} + V_{1j} X_{ij} + r_{ij}) \tag{3}$$

Fitting the model (typically using REML) partitions total outcome variance into *within-school* and *between-school* components. Analyzing variance proportions across random effects (V_{0j} and V_{1j}) and residual error (r_{ij}) reveals where significant variance resides. The estimated coefficients γ_{01}, γ_{10}, and γ_{11} indicate effect sizes of individual and group characteristics on outcomes.

Takeaway: Multilevel models capture the nested structure of hierarchical and multilevel data and determine how levels and variables interact and affect outcome variables.

3 Datasets

For our study of Internet performance variation, we employ *two* datasets. One includes our outcome variables of interest. Both include theoretically salient independent variables that may be associated to the outcomes.

Ookla Speedtest. The main dataset for our analysis consists of an Ookla Speedtest [28] raw extract containing data for speedtests conducted between January 2022 and May 2023 by devices located in 37 urban-core areas of the United States of America (see Figure 1). This private dataset was made available to us through a data use agreement with Ookla and is substantially more fine-grained and detailed than the public use Ookla for Good open datasets [29]. The open datasets aggregate and summarize speedtest results across the globe in quadtiles (mercator tiles of approximately 610.8 meters by 610.8 meters at the equator). In contrast, the private dataset presents both disaggregated results (e.g., throughput, latency, jitter) and metadata (e.g., exact location, date and time, provider, connection type, Ookla server, etc) for each individual speedtest in selected areas. The selected areas were defined by Ookla based on US city boundaries. These were Atlanta GA, Austin TX, Bakersfield CA, Baltimore MD, Boston MA, Chicago IL, Colorado Springs CO, Denver CO, Detroit MI, El Paso TX, Fort Worth & Dallas TX, Fresno CA, Houston TX, Indianapolis IN, Kansas City KS, Long Beach CA, Louisville KY, Memphis TN, Miami FL, Minneapolis MN, New York NY, Oklahoma City OK, Omaha NE, Philadelphia PA, Phoenix & Mesa AZ, Portland OR, Raleigh NC, Sacramento CA, San Antonio TX, San Diego CA, San Francisco & San Jose CA, Seattle WA, Tucson AZ, Tulsa OK, Virginia Beach VA, Washington DC, and Wichita KS. Tests geolocated within those boundaries were included in the dataset, whereas tests outside were not shared with us.



Figure 1: Geographical distribution of speedtests in the Ookla dataset extract.

The variables relevant to our study and available in the Ookla dataset can be classified into three groups: outcome, location, and explanatory. Regarding outcome variables, speedtests measure four main performance metrics—throughput, latency, jitter, and packet loss rate. As discussed in the introduction, our analysis focuses on the latency (idle and under load) and jitter metrics. In this context, latency is defined as the total elapsed time (in milliseconds) between a client device sending an ICMP Echo Request to the Ookla server and the same client receiving the respective ICMP reply from the server. Latency is measured during an idle stage of the speedtest (i.e., without any concurrent test traffic) and individually during the download and upload stages (i.e., latency under load). Since multiple latency probes are sent at each stage (idle, download, and upload), Ookla provides the minimum, maximum, and interquartile mean for each latency metric. Jitter is the mean absolute difference of consecutive latency samples collected during a ping test. Ookla speedtests measure jitter only during the idle stage. In total, we have ten outcome variables of interest (three for each of the three latency measures, plus jitter).

Regarding location, three main variables are of interest: the latitude of the physical location of the user device running the test, the longitude of that location, and the method used to geocode the location. Device locations are geocoded by either GPS or the IP-based geolocation API Maxmind. Finally, regarding pertinent explanatory variables related to our analysis, Ookla provides information on the type of device connection (e.g., Wi-Fi, Ethernet, or Unknown), the Internet provider serving the testing device, the speedtest server participating in the test, the geographical distance in kilometers between testing device and server, and the date and time the test was conducted. We note that additional metadata fields are present in the dataset (e.g., Wi-Fi RSSI signal strength), however, they have high frequency of missing data (e.g., over 90% missing). Consequently, we do not include these fields in this study.

The dataset contains 76 414 601 records. We performed significant preprocessing to clean and filter the raw data. We remove 6 398 173 speedtests that were conducted from devices connected to Virtual Private Network (VPN) services, since latency tests in this case do not measure the direct path from testing device to speedtest

server. We also filter out 1 146 537 records for speedtest servers located outside of the US to control for international/intercontinental effects on latency. Furthermore, we discard 29 852 records with missing provider information and 2229 records with bogus timestamps (i.e., missing or outside the extract window). Some records fail to meet more than one of the previous criteria. After the aforementioned steps, the cleaned dataset contains data for 69 056 767 speedtests. In terms of geographic distribution, this dataset covers 371 191 urban-core census blocks, 32 816 census block groups, 12 177 census tracts, 91 counties, and 28 US continental states. Represented in the dataset are about 18.9 million user devices, 7576 Internet service providers, and 2129 Ookla Speedtest servers.

Neighborhood Disadvantage Index. To capture differences in the level of socioeconomic deprivation across U.S. neighborhoods, we follow a long line of social science research on how neighborhoods affect social processes and generate a neighborhood disadvantage index (NDI). The NDI is based on U.S. census block-group-level population characteristics encompassing residents' economic status, educational attainment, and household structure. These block-group-level structural characteristics tend to be very highly correlated with one another. As such, neighborhood researchers often opt to use these variables to construct a single NDI covariate instead of including all of these census-derived proxies for neighborhood deprivation simultaneously as covariates in multivariate models. Doing so helps reduce dimensionality and mitigate multicollinearity. Concerns about imprecise, unstable, and unreliable estimates of predictors' effects are thus reduced [14, 37, 38, 50, 51]. The single NDI covariate is typically generated by running a principal component analysis (PCA) based on some combination of five to fifteen variables capturing residents' race, class, and household structure characteristics, and then extracting the first component from the PCA to represent neighborhood socioeconomic deprivation. All other components of the PCA are, by definition, orthogonal to this first component and therefore capture neighborhood variance unrelated to socioeconomic deprivation. Although these other sources of neighborhood variance could plausibly predict spatial differences in internet performance as well, they are not well-theorized; in contrast, the socioeconomic deprivation construct is central to theoretical accounts and empirical examinations of spatial disparities in social science research [16].

Our NDI measure is based on five variables commonly included in prior studies' PCA-generated NDI measures: percentage of residents over age 25 with at least a bachelor's degree (Census variable A12002_005), unemployment rate among civilian residents age 16 and over (A17005_003), percentage of households living below the federal poverty level (A13002_002), percentage of households that received public assistance in the past 12 months (B19057_002), and percentage of female-headed households with children but no spouse present (B11001_006). We note that the very high correlations among these variables and other variables occasionally included (e.g., those related to race/ethnicity) in the PCA process that generates NDI measures ensures that as long as multiple neighborhood sociodemographic characteristics are included in the PCA, the NDI's value is not highly sensitive to the inclusion or exclusion of specific variables [19].

Table 1 summarizes the results of the principal component analysis (PCA). We use the first principal component (PC1) as our neighborhood disadvantage index for three main reasons. First, PC1 is the only component that satisfies both the scree test and the Kaiser criterion [5, 15, 21]. Its eigenvalue exceeds not only the standard Kaiser criterion threshold of 1 but also the more stringent threshold of 2, whereas the remaining components have much smaller and tightly clustered eigenvalues (0.55-0.86), each explaining only 11-17% of the variance. The sharp drop in eigenvalues after PC1 indicates that subsequent components contribute only minimal additional information, and the scree test encourages dropping such components.

Second, PC1 strongly loads on the five socioeconomic indicators used in the PCA, i.e., percent bachelor's degree attainment (A12002_005), poverty (A17005_003), unemployment (A13002_002), public assistance (B19057_002), and female-headed households (B11001_006), with loadings of -0.43, 0.39, 0.53, 0.35, and 0.51, respectively. These values are large in magnitude and align with theoretical expectations: higher educational attainment is associated with less disadvantage, while higher poverty, unemployment, public assistance use, and female-headed households correspond to greater disadvantage.

Third, the remaining components (PC2-5) do not capture socioeconomic deprivation. Their loadings are either weakly correlated with the neighborhood indicators or conflict with theory, suggesting that they reflect orthogonal sources of variation unrelated to disadvantage. For example, PC2-4 have positive loadings for bachelor's degree attainment, while negative loadings would be expected to capture disadvantage. Similarly, PC5 is only weakly correlated with three variables (A12002_005, A17005_003, and B19057_002) and the loading for unemployment (A13002_002) conflicts with deprivation theory. As a robustness check, we re-estimated our main models using the five individual socioeconomic variables rather than the composite index (PC1, standardized to mean 0 and SD 1, with higher values indicating greater disadvantage). These models produced substantively indistinguishable results, supporting our use of PC1 as a parsimonious and theoretically coherent measure of neighborhood structural disadvantage.

Takeaway: We use a non-public, disaggregated Speedtest dataset from Ookla for the outcome variable results and metadata about the tests and infrastructure. The American Community Survey data allow us to bring demographic variables into our models and evaluate their associations with the Ookla-derived outcome variable.

4 Multilevel Modeling of Internet Access Performance

As described in Section 2, multilevel models allow variance in a given outcome to be decomposed into components attributable to each layer or unit within a hierarchical structure. This decomposition helps reveal the levels at which a disproportionate amount of variance in the outcome resides, pointing the way to the types of interventions that might be most impactful in improving levels of, and reducing inequalities in, Internet performance.

In this section, we present our approach for multilevel modeling, using crowdsourced Internet performance outcomes. Briefly stated, our approach consists of three major steps: (§4.1) sample definition,

Component	Eigenvalue	Variance Proportion	Variable Loadings				
			A12002_005	A17005_003	A13002_002	B19057_002	B11001_006
1	2.075	41.5%	-0.433	0.394	0.527	0.350	0.506
2	0.863	17.3%	0.464	0.199	-0.133	0.831	-0.193
3	0.818	16.4%	0.322	0.843	0.034	-0.407	-0.136
4	0.689	13.8%	0.701	-0.265	0.352	-0.146	0.541
5	0.554	11.1%	-0.042	0.154	-0.761	0.011	0.629

Table 1: Results from the Neighborhood Disadvantage Index (NDI) principal component analysis (PCA).

(§4.2) unconditional multilevel modeling, and (§4.3) conditional multilevel modeling.

4.1 Sample Definition

Given the cleaned dataset as described in Section 3, we focus our analysis on three subsamples: A, B, and C. In the next paragraphs, we describe each of the subsamples, presenting the selection criteria for their contents and the rationale behind these criteria.

The first subsample, Sample A, consists of all speedtests with values simultaneously valid for both the minimum idle latency (*idle_latency_min*) and the *jitter* outcome variables conducted from devices connected via Ethernet and geocoded using GPS. In our exploration of the geocoding location data, we observe that the available IP-based geolocation via Maxmind is precise only down to the ZIP Code area level, with points usually placed at the area centroids. Given our goal of studying spatial variation at geographical levels more fine-grained than ZIP Codes (e.g., Census Blocks and Blocks Groups). All three selected subsamples include only GPS-located tests, which enable the necessary degree of spatial precision. Still, we note that cross-testing considering IP-geolocation subsamples support our GPS findings, though limited to higher geographical levels.

Regarding device connection type, since our study centers around latency measures, in Sample A, we prioritize selecting for analysis those speedtests that were conducted via an Ethernet connection as opposed to Wi-Fi tests, which are subject to additional uncertainty factors. Challenging this prioritization, we found that all outcome variables, except for *idle_latency_min* and *jitter*, present high missing rate in Ethernet tests. An investigation into this fact reveals that the other outcome variables only started being collected in May 2022. Further, collection appears to be rolled out progressively, with many tests conducted after May 2022 still missing those variables. Table 2 shows the number of valid values for each outcome variable given the connection type, and motivates our two remaining samples. We note that Maxmind-geolocated tests suffer from the same problem (low number of complete Ethernet tests), with an additional challenge that for about 32 million of those tests (46% of all speedtests) metadata regarding connection type (Ethernet or Wi-Fi) is missing.

Sample B consists of all GPS-geocoded speedtests with simultaneously valid values for both the *idle_latency_min* and *jitter* outcome variables conducted from devices connected either via Ethernet or Wi-Fi. The purpose of this subsample is to estimate the effects of Wi-Fi on the two outcomes that are widely available for both types of connection. The analysis of these effects will help us frame the results of the models fitted over our third subsample.

Sample C consists of all GPS-geocoded speedtests with simultaneously valid values for all outcome variables of interest to our study from Wi-Fi connected devices. We select this subsample for analysis due to the extremely low rate of availability of latency-under-load outcome variables for Ethernet speedtests (only about 12 thousand valid records). Table 3 summarizes our samples and their geographical coverage. Despite the expected added noise due to Wi-Fi communication, together with Samples A and B, this third sample can help us appropriately gauge the effects of spatial, technological, and sociodemographic characteristics on latency. We note that it brings both the highest availability for outcome variables and the greatest geographical coverage.

4.2 Unconditional Multilevel Modeling

For each selected sample, we start our modeling by fitting an individual unconditional multilevel model on each of the latency outcome variables. Each of these initial models only includes the specific outcome variable and the variables that assign observations to "groups" at each one of the levels of interest. No explanatory variables are included; these will be added into subsequent (conditional) models (§4.3).

In our study, we consider seven distinct levels: Test, Device, (Census) Block, (Census) Block Group, (Census) Tract, County, and State. The description for each level is presented in Table 4. At the lowest level of our seven levels of interest, *Test*, we seek to capture the variance in results *between tests* conducted from the same user device while situated within a single census block. Another name for this level is *within devices*. At the *Device* level, we look for the variance *between devices* when conducting tests from the same census block. An alternative name for this level is *within census blocks*. The remaining levels all correspond to well-known spatial boundaries used in research—namely census block, block group, tract, county and state—and capture the variance between entities from the respective level name. For example, the *County* level captures variance between counties located in the same state. We determine grouping within each of the seven levels by executing a spatial join between test locations (as geocoded in latitude and longitude coordinates by GPS) and US Census TIGER/Line boundaries [4].

The results of this initial unconditional model reveal the levels at which meaningful variance in the latency outcome can be observed. When fitted, an unconditional model estimates the variance at each level. We determine the levels of meaningful variance by calculating the Intraclass Correlation Coefficient (ICC) for each level. The ICC

	GPS			Maxmind		
Outcome Variable	**Wi-Fi**	**Ethernet**	**Unknown**	**Wi-Fi**	**Ethernet**	**Unknown**
idle_latency_min	12801307	600404	36649	5123398	1585171	32271095
idle_latency_iqm	9295545	12293	0	2725520	58133	24905496
idle_latency_max	9295545	12293	0	2725520	58133	24905496
download_latency_min	9233466	12259	0	2635306	12531	24905472
download_latency_iqm	9233466	12259	0	2635306	12531	24905498
download_latency_max	9233466	12259	0	2635306	12531	24905498
upload_latency_min	9266937	12270	0	2655218	12576	18506044
upload_latency_iqm	9266937	12270	0	2655218	12576	18506075
upload_latency_max	9266937	12270	0	2655218	12576	18506075
jitter	12765900	592002	32202	5109523	1573159	32248558
Number of Tests	12801307	600404	36649	5123398	1585171	32271095

Table 2: Number of non-missing values for outcome variables given geocoding method (GPS or Maxmind) and connection type (Wi-Fi, Ethernet, or Unknown).

	Sample A	**Sample B**	**Sample C**
Tests	459382	10695096	7556356
Devices	96608	2383801	1713861
Blocks	46453	322590	295521
Block Groups	20572	32360	32106
Tracts	9916	12012	11918
Counties	56	79	74
States	25	27	26

Table 3: Summary of number of tests and devices and geographical coverage for the selected samples.

for a level i is calculated as the ratio between the variance at level i and the sum of variance at all levels (Eq. 4).

$$ICC_i = \frac{\sigma_i^2}{\sum_{j \in \text{Levels}} \sigma_j^2} \tag{4}$$

Evaluating level ICC is important not only for analytical insight but also for the modeling process. The elimination of levels at which minimal variance resides helps simplify both fitting for subsequent (conditional) models and the interpretation of their results. Complex models with extensive hierarchies are often difficult to fit, leading to fit convergence problems. This also informs the explanatory-variable selection process, enabling us to focus on variables that are expected to vary at the most expressive levels (e.g., available providers and neighborhood disadvantage index).

Another possible source of convergence problems is the sample distribution of the outcome variables. Indeed, we find that multilevel models are unable to converge on raw latency measures in our dataset. This is because multilevel models generally expect the outcome to follow a distribution resembling the normal distribution, whereas latency measures are known to present long-tailed distributions [9]. To address this challenge, we apply the well-known log-transform (i.e., $log(x + 1)$ where x is the outcome variable) and subsequently standardize values for all outcome variables before fitting [10]. As expected, we find this transformation effective, enabling model convergence for all samples.

4.3 Conditional Multilevel Modeling

Once the levels explaining the majority of variance in the outcome have been selected, the potentially explaining variables are then introduced to each unconditional model to form a conditional model. Multilevel models accept both numerical and categorical data as input variables. Categorical variables are transformed with one-hot encoding (as illustrated in Table 5). Numerical variables that are expected to have a linear effect on the outcome can be directly introduced to the model, without transformations. Numerical variables with non-linear effects can be split into multiple ranges and then treated as categorical variables.

After fitting the conditional model, a coefficient and a p-value is calculated for each input variable. The p-value indicates whether the null hypothesis (i.e., the variable has no effect on the outcome) can be rejected, while the coefficient estimates the effect that the variable is expected to have on the outcome, net of all other explaining variables' effects. We note that, since in our case we apply the log transform on the outcome variables, the coefficient estimates represent a multiplicative effect on the outcome, as opposed to a direct additive effect. For hypothesis testing, we work with a 95% confidence level. The results of the conditional multilevel model can be used to determine the statistical significance and magnitude of each variable's effect on the outcome, while accounting for the spatial clustering of the error term structure. In other words, the conditional model output can help identify the characteristics of the device conducting a speedtest, of the Internet connection being used by the device, of the ISP offering the connection, of the area where the device is located, of the speedtest server the test is being conducted against, and so on that have significant effects on the latency outcome.

Regarding the explanatory variables, for Samples A and C we initially focus on the neighborhood disadvantage index (NDI) to understand the effect of sociodemographics on latency measures.

Level	Description
Test	Captures the variation between tests conducted from the same user device within the same census block.
Device	Variation between devices running tests within the same block, not captured by the previous level.
Block	Variation between census blocks within the same block group, not captured by previous levels.
Block Group	Variation between census block groups within the same tract, not captured by previous levels.
Tract	Variation between census tracts within the same county, not captured by previous levels.
County	Variation between counties within the same state, not captured by previous levels.
State	Variation between US states, not captured by any of the previous levels.

Table 4: Hierarchical levels considered for our multilevel modeling of Internet access performance, from the lowest to the highest level.

	ISP
0	IS&P
1	Ynfinity
2	IS&P
3	Horizon
4	Horizon
...	...

(a) Before

→

	isp_0	isp_1	isp_2	...
0	**1**	0	0	...
1	0	**1**	0	...
2	**1**	0	0	...
3	0	0	**1**	...
4	0	0	**1**	...
...	...	...	...	...

(b) After

Table 5: One-hot encoding transformation method example. Each unique provider name is mapped to a boolean membership variable. Each row has value 1 (true) for the membership variable respective to its original value and 0 (false) for all others.

Subsequently, we also explore the technical variables available in the Ookla dataset. Namely, the hour of the day when the measurement was taken (i.e., 0-23), the day of the week (i.e., Sunday to Saturday), the running week of our sample (e.g., 45th week of 2022), the running month (e.g., May 2022), the speedtest server responding to the user device, the distance between user device and server (in kilometers), and the Internet service provider of the device. For Sample B, we only consider the connection type (Ethernet or Wi-Fi) to evaluate the effect of Wi-Fi on measured latency.

One challenge we face with the conditional models is the large number of distinct servers and providers in the dataset (see §3). This constitutes a challenge because each unique value introduces an additional (dummy) variable that has to be fitted during optimization. Paired with the large number of observations in our samples, this leads to convergence problems and fitting errors. To address this challenge, we limit our samples to tests from the top-n providers and against the top-m servers. The top-n providers are defined as a set of the n providers with the largest number of individual tests. The top-m servers set definition follows the same logic. In our study, we use $n = 50$ and $m = 200$ as they maximize the geographical coverage while still allowing for successful model convergence. Limiting to the top-50 providers and top-200 servers keeps coverage to about 87%, 92%, and 92% of the census blocks represented in the original Sample A, B, and C, respectively. To conclude, we note that we cross-tested samples with unconditional modeling and found equivalent results. Further, the summary depicted in Table 3 represents coverage after restricting the original samples to the top providers and servers.

Takeaway: Our multilevel modeling approach to Internet access performance consists of three interactive steps: (1) Sample definition, where we select specific samples to focus our analysis; (2) Unconditional modeling, where we determine the levels at which a substantial portion of the outcome's variance reside; (3) Conditional modeling, where we use metadata variables to evaluate how well they can explain the outcome and its variance estimated at each level of the multilevel model.

5 Results and Discussion

In this section, we present the results of our study and discuss their implications. For multilevel modeling, we utilized the R (version 4.3.3) language lmer function defined in the lme4 library (version 1.1.35.3). The fitting optimization uses the nloptwrap algorithm with a maximum of 10 000 iterations. To improve computational efficiency and convergence, derivative calculations are disabled (calc.derivs = FALSE) and relax tolerances (tol = 2e-3). We ran more than a hundred total model fitting calls considering distinct combinations of samples, outcome variables, levels, and explanatory variables. This required more than 900 computing hours on a Debian GNU/Linux 12 (5.15.0-97-generic) server configured with a 40-core Intel(R) Xeon(R) CPU E5-2660 v3 @ 2.60GHz and 202 GB of RAM. We begin by reporting on our unconditional multilevel modeling analysis, and conclude with the conditional modeling.

5.1 Multilevel Characterization of Latency Variation

As described in Section 4, unconditional multilevel models help us determine the levels at which there is meaningful variance in our outcome of interest. The variance estimates directly answer our first research question (RQ1 in §1), describing the proportion of the variance (ICC) in Internet performance residing between and within US states, counties, states, census tracts, census block groups, and census blocks. Here we present and discuss the unconditional model results for Samples A and C.

Sample A. Table 6 depicts unconditional results for both the *idle_latency_min* and *jitter* outcome variables. The table shows the absolute amount of variance (and the percentage of the total variance, i.e., ICC) attributed to each level by the fitted model. In other words, each row represents how much of the total variance

Levels	Idle Latency (min)	Jitter
Test	0.16 (16.19%)	0.43 (46.76%)
Device	0.45 (44.89%)	0.33 (35.9%)
Block	0.12 (12.01%)	0.06 (6.48%)
Block Group	0.04 (3.88%)	0.03 (3.65%)
Tract	0.07 (6.92%)	0.04 (4.52%)
County	0.15 (14.64%)	0.02 (2.64%)
State	0.01 (1.48%)	0.00 (0.04%)
Total	1.01	0.91

Table 6: Variance and ICC estimated for each level for each outcome variable in the Sample A.

can be attributed to differences between groups within the same level while also considering the differences between groups at all other levels.

From Table 6, we first observe that multiple levels have a significant share of the total variance apportioned to them. In other words, there is substantial variation in the outcome between groups (or individuals) at those levels. This confirms our expectation that latency measures are well-suited for a multilevel modeling approach. For *idle_latency_min*, about 45% of the total variance resides between devices within census blocks, while another 16% resides between tests, 15% between counties, and 12% between blocks. In other words, we observe the *idle_latency_min* variance to be concentrated mostly at very fine geographical levels and the county (urban-core) level. For *jitter*, about 83% of the total variance resides between tests and devices. This suggests that device and network characteristics, as well as activity during testing may influence jitter measures more strongly than neighborhood or geographical characteristics.

Sample C. Table 7 presents unconditional results for this sample. We observe a very similar pattern in the share of variance attributed to each level. However, in outcome variables other than *idle_latency_min*, the ICC for the county level is generally a few percentage points lower than for Sample A. Further, for all outcome variables, the variance estimated to reside between tests dominates all other levels. Both of these deviations from the pattern observed for Sample A are expected due to the noise added to latency measures by wireless communication over Wi-Fi. This is especially to be expected during the timeline of the Ookla dataset, when market research estimated households to have about 22 connected devices on average [2, 11], double the pre-pandemic average.

Looking at Sample C by itself, we also observe ICC for the block level to be consistently between 3% and 7% across all metrics and versions (min, iqm, and max). This suggests a steady effect of census block characteristics on latency and jitter measures. For latency under load metrics and the county level, the ICC is more expressive for the minimum version of each metric than the IQM and maximum, which can get quite low. This result seems to indicate that county characteristics have a stable effect in determining the best case scenario latency (min) while having a subtler effect for the other cases.

All of these results together lead us to prioritize county- and census block-level explanatory variables, as well as within-device temporal differences, in our efforts to explain as much of the total outcome variance as possible. Further, as previously discussed in Section 4.2 and revealed in the results, should subsequent *conditional* models need to be simplified to facilitate fitting convergence, the test, device, census block, and county levels warrant prioritization.

Takeaway: Our unconditional multilevel analysis of Internet access latency and jitter reveals that variance follows a spatially nested structure and is due primarily to differences between longitudinal measurements (test) and devices, while differences between counties and census blocks explain a modest share of the variance.

5.2 Effect of Explanatory Variables on Latency Measures

We build on the unconditional analysis by fitting *conditional* multilevel models that reveal the effects of selected explanatory variables on the outcome variables, while also taking the hierarchical structure of variation into consideration. The conditional models help us answer our second research question (RQ2 in §1), describing the proportion of the measured variance that can be explained by sociodemographics or other factors.

Connection Type: Ethernet vs Wi-Fi. We start by evaluating the effect of connection type (Ethernet vs Wi-Fi) on latency to help contextualize the subsequent results and discussion for the remaining explanatory variables. We estimate the effect of Wi-Fi on the *idle_latency_min* and *jitter* outcomes in Sample B. For both outcome variables, the effect estimate is found to be statistically significant with near-zero p-values ($<1 \times 10^{-9}$). The effect of measuring latency on a Wi-Fi connection is estimated to increase the measured value by about 9.72% for *idle_latency_min* and 25.42% for *jitter*, a meaningful degradation in both cases. Note that these effects are multiplicative due to the application of the log-transformation to the outcome values. These results fit with expectations that Wi-Fi has generally higher base latency and jitter than Ethernet. They also help explain the growth in ICC for the lowest level (test) between Samples A (Table 6) and C (Table 7).

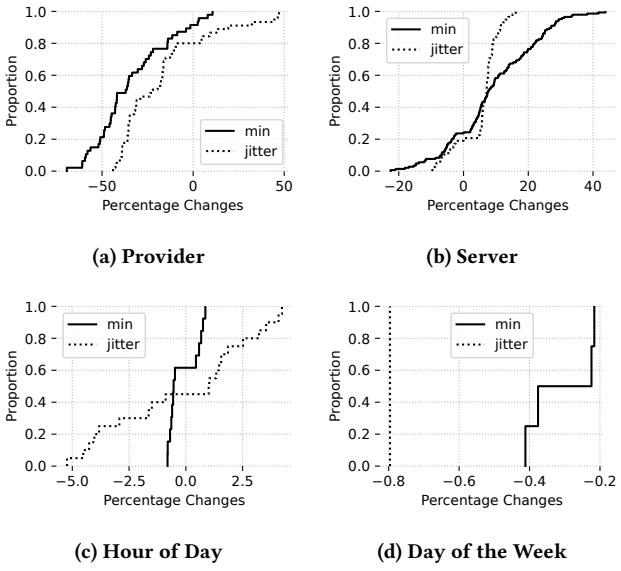
Neighborhood Disadvantage Index. We evaluate NDI to help us understand what proportion of the measured outcome variance that exists is explained by neighborhood sociodemographic composition (RQ2 in §1). We fit conditional models including NDI as an explanatory variable and report the results for Samples A and C in Table 8. For all evaluated outcome variables, the effect is found to be statistically significant, with maximum p-value at 3.74e-11 for *upload_latency_iqm*. The magnitude of this effect, however, is small in all cases, below 1.5% variation in the outcome for each standard deviation variation in the NDI score. For all evaluated outcomes, the effect represents an increase, except for *upload_latency_iqm*, where a 0.19% reduction is estimated. The effects also appear smaller in latency under load contexts than for idle latency. Nevertheless, all results suggest that neighborhood disadvantage plays quite a small role in estimating Internet performance in relationship to latency and jitter outcomes.

Levels	Idle Latency min	Idle Latency iqm	Download Latency min	Download Latency iqm	Download Latency max	Upload Latency min	Upload Latency iqm	Upload Latency max	Jitter
Test	0.28 (41.72%)	0.35 (56.07%)	0.50 (57.31%)	0.45 (50.27%)	0.54 (59.27%)	0.37 (47.48%)	0.45 (50.29%)	0.52 (56.58%)	0.56 (76.34%)
Device	0.21 (30.3%)	0.15 (23.55%)	0.25 (28.69%)	0.35 (39.26%)	0.30 (33.37%)	0.26 (33.54%)	0.37 (40.68%)	0.33 (36.04%)	0.13 (17.91%)
Block	0.05 (7.32%)	0.04 (6.34%)	0.04 (4.36%)	0.05 (5.19%)	0.03 (3.83%)	0.04 (4.59%)	0.04 (4.68%)	0.04 (3.89%)	0.02 (3.07%)
B. Group	0.01 (1.71%)	0.01 (1.51%)	0.01 (0.8%)	0.01 (0.75%)	0.01 (0.62%)	0.01 (0.93%)	0.01 (0.77%)	0.01 (0.73%)	0.00 (0.66%)
Tract	0.03 (4.29%)	0.02 (3.48%)	0.01 (1.7%)	0.02 (2.23%)	0.01 (1.36%)	0.01 (1.26%)	0.01 (0.67%)	0.01 (0.75%)	0.01 (1.18%)
County	0.09 (13.41%)	0.05 (8.62%)	0.04 (5.14%)	0.02 (1.85%)	0.01 (1.3%)	0.07 (8.42%)	0.01 (1.01%)	0.00 (0.27%)	0.00 (0.65%)
State	0.01 (1.24%)	0.00 (0.43%)	0.02 (1.99%)	0.00 (0.44%)	0.00 (0.25%)	0.03 (3.78%)	0.02 (1.89%)	0.02 (1.75%)	0.00 (0.19%)
Total	0.68	0.62	0.87	0.90	0.90	0.79	0.90	0.93	0.74

Table 7: Variance and ICC estimated for each level for each outcome variable in the Sample C.

Outcome Variable	Sample A	Sample C
idle_latency_min	0.6914%	1.1460%
idle_latency_iqm	-	1.1979%
idle_latency_max	-	1.2663%
download_latency_min	-	0.5169%
download_latency_iqm	-	0.4590%
download_latency_max	-	0.36715
upload_latency_min	-	0.4999%
upload_latency_iqm	-	-0.1963%
upload_latency_max	-	0.2239%
jitter	0.7715%	1.4032%

Table 8: NDI effect estimates on the outcome variables. All estimates are statistically significant but small in magnitude. A hyphen indicates that the outcome variable is not present in the sample.



(a) Provider **(b) Server**

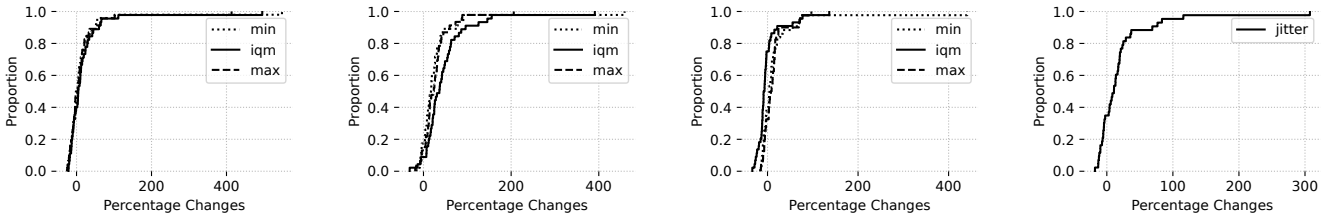
(c) Hour of Day **(d) Day of the Week**

Figure 2: Categorical explanatory variables statistically-significant effect estimates on Sample A outcome variables.

Internet Service Provider. Following the results for NDI, we look at the remainder of the explanatory variables to answer the second part of RQ2 (§1). An analysis of the effect that the ISP connecting the testing device to the Internet is justified by the fact that several choices made by ISPs can have direct impacts on latency measures. Some crucial examples are how individual ISPs choose to deploy their infrastructure and to interconnect with others, as well as the particular technologies they employ to offer Internet access. Although we cannot directly test these possibilities, or other plausible explanations for ISP effects on latency, due to data constraints, it is still interesting to evaluate whether simply knowing the ISP helps understand spatial latency variation. We present statistically-significant results for this modeling analysis in Figures 2a and 3.

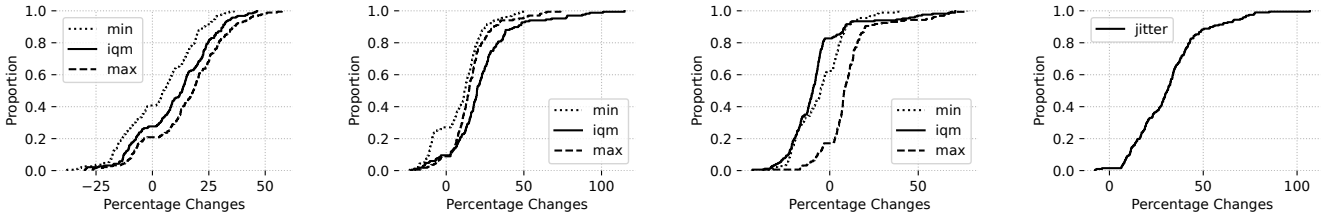
For both samples, for all outcome variables of interest, at least 41 out of the 50 (82%) evaluated ISPs present statistically significant estimated effects (p-value < 0.05). Further, we observe that the effect of ISP on latency measures is quite large. For Sample A, for example, the estimated effect on *idle_latency_min* can be up to a 69% decrease (or difference) when comparing two distinct ISPs. For jitter, this number can also be up to about 50%. For Sample C, we find the that the estimated effect can be as large as 500% in the case of both idle and latency under load, while as large as 300% for jitter. Comparing the different versions (min, iqm, and max) of the latency metrics, we observe small differences on the size of the effect, pointing to a consistent effect. Further analysis reveals that the effect represents the difference between access provided by a fiber optic ISP and a GEO-satellite ISP. For a LEO-satellite ISP, the effect is estimated at 101.5%. These results suggest that the provider to which a testing device is connected has strong influence on the expected measured latency. Moreover, these results reinforce the importance of mapping ISP availability coverage, and beyond that, ISP adoption across geographical areas in the United States.

Speedtest Server. Continuing the analysis, we explore the effect that testing against different speedtest servers has on the latency outcomes. The reason for investigating this is that speedtest server deployment does not follow a single standard; rather, servers are hosted by sponsoring organizations (including, sometimes, ISPs themselves) using their own hardware and infrastructure. As such, some variation can be expected in terms of server performance and connectivity. We evaluate the effect of speedtest servers and report statistically-significant coefficients in Figures 2b and 4. At a minimum (Sample A and *jitter*) 54 out of the 200 (27%) evaluated



(a) Idle Latency **(b) Download Latency** **(c) Upload Latency** **(d) Jitter**

Figure 3: Provider category statistically-significant effect estimates on the outcome variables.



(a) Idle Latency **(b) Download Latency** **(c) Upload Latency** **(d) Jitter**

Figure 4: Server category statistically-significant effect estimates on the outcome variables.

Outcome Variable	Sample A	Sample C
idle_latency_min	0.0040%	0.0254%
idle_latency_iqm	-	0.0213%
idle_latency_max	-	0.0167%
download_latency_min	-	0.0159%
download_latency_iqm	-	0.0117%
download_latency_max	-	0.0046%
upload_latency_min	-	0.0277%
upload_latency_iqm	-	0.0190%
upload_latency_max	-	0.0000%
jitter	0.0052%	-0.0065%

Table 9: Server distance effect estimates on the outcome variables. All estimates are statistically significant but infinitesimal in magnitude. A hyphen indicates that the outcome variable is not present in the sample.

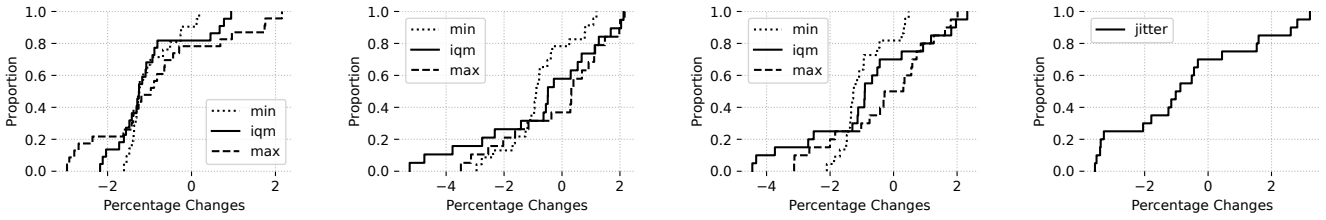
servers present statistically-significant effects. At a maximum (Sample C and *jitter*) all but two of the evaluated servers (99%) have statistically-significant effects.

From the plots, we observe that the magnitude of the effect is slightly smaller than the ISP effects. Nevertheless, the server effects are still quite large, with maximum absolute effect at 43.85% for Sample A with *idle_latency_min* and over 100% for Sample C with *download_latency_iqm* and *jitter*. Considering that, by default, speedtests are conducted against the most geographically proximate speedtest server, this result suggests that there are considerable resource differences between servers (or the individual paths to servers) deployed in each of the urban cores analyzed in our dataset.

Considering the different versions of the latency metrics, we observe modest differences on the size of the effect for idle latency and download latency. For upload latency, the differences appear more clearly when comparing *max* to *min* and *iqm*, although relatively still small.

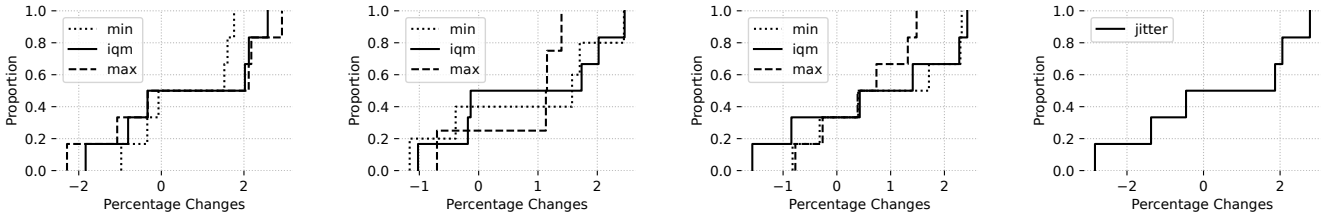
Given the apparent importance of the server in explaining our outcome variance, we next evaluate whether latency differences may reflect the distance between the individual device and the server used. From Table 9, we observe that despite the estimates being statistically significant, they are also infinitesimal. The largest absolute effect is estimated at only 0.0277% for each additional kilometer of distance. Consequently, differences in performance or quality of access between servers appears to matter more than distance between testing device and server. These results underscore the importance of identifying the servers participating in speedtests for certain areas, as well as ensuring analytic samples of speedtests encompass a wide array of servers.

Moment in Time. One final point of interest is whether knowing the precise moment when a latency measurement took place explains the considerable amount of variance detected between measurements drawn from the same device. It is plausible that utilization and congestion levels at different times of the day or specific days (e.g., workday versus weekend versus holiday) could influence the measured latency outcome. As mentioned previously, we track and evaluate the effect of the hour of the day (*hod*) when the measurement was taken, the day of the week (*wd*), the running week of our sample (*yw*), and the running month (*ym*). From this analysis, we find several hours of the day (*hod*) and days of the week (*wd*) with statistically significant estimated effects for Samples A and C. We do not find, however, any particular week or



(a) Idle Latency **(b) Download Latency** **(c) Upload Latency** **(d) Jitter**

Figure 5: Hour of day statistically-significant effect estimates on the outcome variables.



(a) Idle Latency **(b) Download Latency** **(c) Upload Latency** **(d) Jitter**

Figure 6: Day of the week statistically-significant effect estimates on the outcome variables.

month in the dataset to have a statistically significant effect on the latency and jitter outcomes. We present the significant effects for Sample A in Figures 2c and 2d and for Sample C in Figures 5 and 6.

For Sample A, the day of the week effect is small, under 1%, similar to NDI effects. The hour of the day effect is similar for *idle_latency_min*, but more expressive for *jitter*, with maximum absolute effect estimated at around a 5% decrease. For Sample C, the effects of both *hod* and *wd* appear overall similar or slightly stronger than in Sample A for all outcome variables. These results suggest that the well-known daily and weekly patterns in Internet usage [27, 34] have a meaningful albeit small effect on latency measures. To conclude, we observe that the effect of *hod* appears slightly more marked in situations under load than when idle. More specifically, the *hod* effect varies from around -5% to 2% for download and upload latency, while it varied from around -3% to 2% for idle latency. This seems to indicate that the sharing of Wi-Fi Internet access (within-households) strengthens the effects of the well-known Internet usage patterns.

Takeaway: Our conditional multilevel analysis indicates that the latency variance observed across levels is mostly explained by differences between Internet service providers and speedtest servers, and to a smaller extent the moment when measures were taken. Surprisingly, neighborhood disadvantage was found to explain only a very small portion of the variance in latency and jitter outcomes.

6 Limitations

Crowdsourced Speedtest Datasets. Crowdsourced speedtest datasets collected by Ookla [28] and Measurement Lab (M-Lab) [22] are subject to self-selection bias. People tend to run speedtests when they sense a problem or irregularity with their Internet connection. Further, middle and high-income households have a noticeably higher participation rate in running speedtests compared to low-income households [40]. These factors may introduce bias to the speedtest datasets in terms of locations where tests originate and network conditions under which they are conducted. In addition, our dataset extract is restricted to speedtests conducted from select urban core locations. These limitations hinder the ability to translate or extend our conclusions to areas beyond those contemplated in our dataset, especially in regards to rural areas.

At a high level, speedtest applications are similar to many other Internet-based applications wherein a user device communicates with an application server deployed on a nearby datacenter or colocation facility. Nevertheless, this speedtest application differs in a few notable ways. First, the content of speedtest traffic is optimized to measure bottleneck capacity at a moment in time, while general application traffic serves functional purposes (e.g., browsing, content streaming, and teleconferencing communication). Second, some ISPs may prioritize or deprioritize speedtest traffic, treating it differently than application data. Third, application traffic benefits from content delivery networks (CDNs), edge caching, and other myriad performance optimizations. Meanwhile, speedtest servers are hosted voluntarily by organizations (usually ISPs) and represent specific test endpoints that may not reflect typical content delivery paths or performance. For example, we observe significant variation in latency and jitter across speedtest servers.

Multilevel Modeling. Multilevel models are useful for partitioning outcome variance among hierarchical levels and estimating the effects of explanatory variables given a hierarchical spatial structure. These models, however, present a few challenges and limitations.

As described in Section 5, all our multilevel models combined took over 900 hours to be fitted. The individual model fitting runtime varied from about 5 minutes for unconditional models for Sample A to several hours for certain conditional models for Sample C. We observe that the computational burden imposed by models grows quickly with the increase in the number of hierarchical levels, distinct groups in each level, explanatory variables (including individual category dummy variables), and total observations. Given the scale and coverage of our samples, in multiple cases, we were unable to fit models directly due to memory and in some cases, computational time constraints. We address this challenge in four ways. First, we configure the fitting optimizer to use an algorithm that requires lower memory usage (nloptwrap) and skip certain computationally heavy procedures (such as derivative calculations) for intermediary steps. Second, we limit the scope of the analysis to the top-50 most popular providers and top-200 most popular servers, according to the number of tests. Third, for Samples B and C, we reduce model complexity by limiting the number of levels to three (test, device, and county). Fourth, for conditional models for Sample C, we further restrict model complexity by evaluating each explanatory variable independently. We highlight that these optimizations and adjustments, despite in some cases constraining our scope, are necessary to make running times and the analysis itself practical.

Takeaway: Crowdsourced speedtest datasets represent imperfect proxies into Internet performance variation subject to self-selection bias and potential differentiated treatment from production traffic. Any attempt to extend or generalize conclusions over these datasets should take these facts into consideration. Multilevel models struggle to scale to massive datasets with several hierarchical levels. This dynamic requires several optimizations and adjustments that constrain the scope of analysis.

7 Related Work

Throughout the years, many research efforts have sought to characterize broadband Internet performance, both in the academic [3, 45] and policy [6–8, 12] communities. The necessity for Internet access in modern society as highlighted by the recent global COVID-19 health crisis has increased the demand for understanding how Internet performance is distributed spatially. That is, where are the communities and neighborhoods whose Internet performance is sub-standard and are in need of intervention?

Given the challenges of collecting a wide sample of Internet performance, the majority of previous work in this area has used crowdsourced datasets collected by Measurement Lab (M-Lab) [22] and Ookla [28] with their respective speed tests. Since speed tests were originally designed to measure the performance of a single access link, these datasets present limitations when applied to spatial modeling. For example, M-Lab does not directly collect the location of devices conducting tests. Consequently, work using M-Lab's datasets tends to focus on aspects other than spatially-based performance [8, 46] or considers IP-based geolocation at much larger geographical scales (e.g., countries, continents). [17, 39]. The Ookla Speedtest dataset does collect GPS-based geolocation. However, in its public version, the dataset is spatially aggregated in quadtiles [12, 26, 31]. This aggregation hides important variation that may occur at the access link or test device level as well as metadata that may help us understand such variation. In line with more recent research [23, 33, 40, 42], in this work we base our analysis on disaggregated test data obtained through a data use agreement with Ookla.

Regarding the outcome of interest, research on broadband access has focused on analyzing speed (or measured throughput) variation [1, 7, 8, 13, 23, 25, 30, 32, 40, 46]. Previous work with such focus has led to important observations on the relationship of demographics, affordability, and Internet adoption. For example, Paul *et al.* [32] have noted that geographical type (i.e., urban or rural) and area income level have the strongest relationship to speeds. Our work adds to a recent shift in also paying attention to latency, given low latency as a key differentiator of current user experience as well as a crucial enabler of future applications (e.g., 360-degree video streaming, autonomous vehicles) [39]. In this context, Høiland-Jørgensen *et al.* [17] investigated the amount of variation in latency experienced by end-users, noting that latency distributions tend to be heavy-tailed. Sharma *et al.* [42] explored ways for identifying areas with consistent latency to establish boundaries for sampling Internet performance.

One notable limitation of previous work that examines both latency and throughput is that this work tends to analyze performance only at a single geographical level. Høiland-Jørgensen *et al.* [17] focused on continents. Others focused on countries [1], states [20], counties [7], census tracts [12, 13, 39, 40], or census block groups [23, 31, 32]. Sharma *et al.* [42] evaluated three distinct boundary representations, while ultimately focusing on a single representation (H3 system) and level (resolution 8 hexagons) [48]. In our work, we note that it is still unknown at what geographical levels variation in Internet performance actually appears, as well as what factors help explain any such variations. To the best of our knowledge, our work is the first exploration aimed at answering these open questions regarding the spatial variation of Internet performance.

Takeaway: Our work is the first to use multilevel modeling to characterize spatial variation in Internet performance. This enables us to determine the spatial scales and associated factors that best explain observed variation in the measured outcomes.

8 Conclusion

In this paper, we addressed two questions that inform efforts aiming to bolster Internet access performance in the United States. Using multilevel models, we estimated the proportion of the variance in Internet performance that resides between and within different spatial scales, including neighborhoods, counties, and states. We also explored what proportion of the variance in performance overall, and at each of these levels, can be explained by sociodemographic and infrastructure factors.

Using latency between Ookla Speedtest user device and server as proxy for Internet performance, we found that a considerable amount of variance in this performance outcome is distributed across all seven levels evaluated. We observed the largest portions of the outcome's variance to reside between tests within devices and between devices themselves, followed to a lesser extent to between census blocks and counties. Conditional multilevel models revealed

that variance in latency is primarily explained by differences among Internet service providers and speedtest servers. Perhaps surprisingly, we found minimal evidence to suggest that community sociodemographic factors (as represented by a commonly-employed neighborhood disadvantage index) help explain an expressive portion of variance in latency. These results challenge widely-held assumptions that community demographics stratify Internet performance across the U.S., and call for more direct measurement of Internet infrastructure (e.g., server and ISP characteristics) to identify particular areas in need of investment.

Acknowledgments

We would like to thank our shepherd and the anonymous reviewers for their valuable feedback. The work was supported by National Science Foundation (NSF) awards CNS-2319603, CNS-2223610, CNS-2213821, and SES-2324515.

9 Ethics

This work does not raise any ethical issues.

References

[1] Dale Alexander and Lika Døhl Diouf. 2022. Illustrating Internet Speed Divides in the Caribbean During COVID-19. *FOCUS Magazine of the Caribbean Development and Cooperation Committee (CDCC)* (12 2022). https://hdl.handle.net/11362/48956

[2] Jana Arbanas, Paul Silverglate, Susanne Hupfer, Jeff Loucks, and Michael Steinhart. 2022. *Mastering the New Digital Life: 2022 Connectivity and Mobile Trends* (3rd ed.). Technical Report. Deloitte Center for Technology, Media & Telecommunications. https://www.deloitte.com/content/dam/assets-shared/legacy/docs/DI_Connectivity-mobile-trends-2022.pdf Survey of 2,005 US consumers conducted in Q1 2022.

[3] Zachary S. Bischof, Fabian E. Bustamante, and Nick Feamster. 2017. Characterizing and Improving the Reliability of Broadband Internet Access. (2017). arXiv:cs.NI/1709.09349 https://arxiv.org/abs/1709.09349

[4] U.S. Census Bureau. 2024. TIGER/Line Shapefiles. https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html. (2024). All legal boundaries and names are as of January 1, 2024. Released September 25, 2024.

[5] Raymond B. Cattell. 1966. The Scree Test For The Number Of Factors. *Multivariate Behavioral Research* 1, 2 (1966), 245–276. https://doi.org/10.1207/s15327906mbr0102_10 arXiv:https://doi.org/10.1207/s15327906mbr0102_10 PMID: 26828106.

[6] Center for Rural Pennsylvania. 2018. Broadband Availability and Access. (2018). https://www.rural.pa.gov/publications/broadband.cfm Accessed November 13, 2024.

[7] Center for Rural Pennsylvania. 2019. *Broadband Availability and Access in Rural Pennsylvania.* Report. Center for Rural Pennsylvania. https://www.rural.pa.gov/download.cfm?file=Resources/PDFs/broadband/Broadband_Availability_and_Access_in_Rural_Pennsylvania_2019_Report.pdf Accessed November 13, 2024.

[8] David D Clark and Sara Wedeman. 2021. Measurement, meaning and purpose: Exploring the M-Lab NDT dataset. In *TPRC49: The 49th Research Conference on Communication, Information and Internet Policy.* https://dx.doi.org/10.2139/ssrn.3898339

[9] David D Clark and Sara Wedeman. 2024. Measurement of Internet access latency: A cross-dataset comparison. In *Proceedings of the TPRC2024 The Research Conference on Communications, Information and Internet Policy.* http://dx.doi.org/10.2139/ssrn.4909679

[10] Douglas Curran-Everett. 2018. Explorations in statistics: the log transformation. *Advances in Physiology Education* 42, 2 (2018), 343–347. https://doi.org/10.1152/advan.00018.2018 arXiv:https://doi.org/10.1152/advan.00018.2018 PMID: 29761718.

[11] Deloitte Center for Technology, Media & Telecommunications. 2023. *Balancing Act: Seeking Just the Right Amount of Digital for a Happy, Healthy Connected Life.* Connectivity and Mobile Trends Survey. Deloitte. https://www.deloitte.com/us/en/insights/industry/telecommunications/connectivity-mobile-trends-survey/2023/connectivity-mobile-trends-survey-full-report.html Fourth edition of Deloitte's Connected Consumer Study.

[12] Peter Cody Fiduccia. 2022. *Deconstructing the Digital Divide: The Geography, Demography, and Spatial Dependence of Internet Stability in the US.* Cornell University. https://www.proquest.com/docview/2724645586?pq-origsite=gscholar&fromopenview=true&sourcetype=Dissertations%20&%20Theses

[13] George S. Ford. 2021. Form 477, Speed-Tests, and the American Broadband User's Experience. *SSRN Electronic Journal* (04 2021). https://doi.org/10.2139/ssrn.3825298

[14] Daniel A. Hackman, Dora Cserbik, Jiu-Chiuan Chen, Kiros Berhane, Bita Minaravesh, Rob McConnell, and Megan M. Herting. 2021. Association of Local Variation in Neighborhood Disadvantage in Metropolitan Areas With Youth Neurocognition and Brain Structure. *JAMA Pediatrics* 175, 8 (08 2021), e210426–e210426. https://doi.org/10.1001/jamapediatrics.2021.0426

[15] J.F. Hair, W.C. Black, B.J. Babin, and R.E. Anderson. 2019. *Multivariate Data Analysis.* Cengage. https://books.google.com/books?id=0R9ZswEACAAJ

[16] Elham Hatef, Xiaomeng Ma, Yahya Shaikh, Hadi Kharrazi, Jonathan P. Weiner, and Darrell J. Gaskin. 2021. Internet Access, Social Risk Factors, and Web-Based Social Support Seeking Behavior: Assessing Correlates of the "Digital Divide" Across Neighborhoods in The State of Maryland. *Journal of Medical Systems* 45, 94 (September 2021). https://doi.org/10.1007/s10916-021-01769-w

[17] Toke Høiland-Jørgensen, Bengt Ahlgren, Per Hurtig, and Anna Brunstrom. 2016. Measuring Latency Variation in the Internet. In *Proceedings of the 12th International on Conference on Emerging Networking EXperiments and Technologies (CoNEXT '16).* Association for Computing Machinery, New York, NY, USA, 473–480. https://doi.org/10.1145/2999572.2999603

[18] J. Hox. 1998. Multilevel Modeling: When and Why. In *Classification, Data Analysis, and Data Highways*, Ingo Balderjahn, Rudolf Mathar, and Martin Schader (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 147–154. https://doi.org/10.1007/978-3-642-72087-1_17

[19] Kahir Jawad, Yana B. Feygin, Michelle Stevenson, Bethany A. Wattles, Jennifer Porter, V. Faye Jones, and Deborah Winders Davis. 2025. The Association between Four Neighborhood Disadvantage Indices and Child Chronic Health Classifications. *Pediatric Research* (May 2025). https://doi.org/10.1038/s41390-025-04143-5

[20] Hanyang Jiang, Henry Shaowu Yuchi, Elizabeth Belding, Ellen Zegura, and Yao Xie. 2023. Mobile Internet Quality Estimation using Self-Tuning Kernel Regression. (2023). arXiv:stat.AP/2311.05641 https://arxiv.org/abs/2311.05641

[21] Henry F. Kaiser. 1960. The Application of Electronic Computers to Factor Analysis. *Educational and Psychological Measurement* 20, 1 (1960), 141–151. https://doi.org/10.1177/001316446002000116 arXiv:https://doi.org/10.1177/001316446002000116

[22] Measurement Lab. 2024. Speedtest by Measurement Lab. (2024). https://speed.measurementlab.net/ Accessed: 2024-12-12.

[23] Hyeongseong Lee, Udit Paul, Arpit Gupta, Elizabeth Belding, and Mengyang Gu. 2023. Analyzing Disparity and Temporal Progression of Internet Quality through Crowdsourced Measurements with Bias-Correction. (2023). arXiv:stat.AP/2310.16136 https://arxiv.org/abs/2310.16136

[24] D.A. Luke. 2019. *Multilevel Modeling.* SAGE Publications. https://books.google.com/books?id=5_vDDwAAQBAJ

[25] Kyle MacMillan, Tarun Mangla, James Saxon, Nicole P. Marwell, and Nick Feamster. 2023. A Comparative Analysis of Ookla Speedtest and Measurement Labs Network Diagnostic Test (NDT7). *Proc. ACM Meas. Anal. Comput. Syst.* 7, 1, Article 19 (March 2023), 26 pages. https://doi.org/10.1145/3579448

[26] Syed Tauhidun Nabi, Zhuowei Wen, Brooke Ritter, and Shaddi Hasan. 2024. Red is Sus: Automated Identification of Low-Quality Service Availability Claims in the US National Broadband Map. In *Proceedings of the 2024 ACM on Internet Measurement Conference (IMC '24).* Association for Computing Machinery, New York, NY, USA, 2–18. https://doi.org/10.1145/3646547.3688441

[27] Office for National Statistics Data Science Campus. 2019. What can internet use tell us about our society and the economy? ONS Data Science Campus Project. (Sept. 2019). https://datasciencecampus.ons.gov.uk/projects/what-can-internet-use-tell-us-about-our-society-and-the-economy/ Extracting social-economic signals from internet traffic data.

[28] Ookla. 2024. Speedtest by Ookla - The Global Broadband Speed Test. (2024). https://www.speedtest.net/ Accessed: 2024-12-12.

[29] Ookla. 2025. Ookla for Good: Open Data. https://www.ookla.com/ookla-for-good/open-data. (2025). Accessed: 2025-11-10.

[30] Udit Paul, Vinothini Gunasekaran, Jiamo Liu, Tejas N. Narechania, Arpit Gupta, and Elizabeth Belding. 2023. Decoding the Divide: Analyzing Disparities in Broadband Plans Offered by Major US ISPs. In *Proceedings of the ACM SIGCOMM 2023 Conference (ACM SIGCOMM '23).* Association for Computing Machinery, New York, NY, USA, 578–591. https://doi.org/10.1145/3603269.3604831

[31] Udit Paul, Jiamo Liu, Vivek Adarsh, Mengyang Gu, Arpit Gupta, and Elizabeth Belding. 2021. Characterizing Performance Inequity Across U.S. Ookla Speedtest Users. (2021). arXiv:cs.NI/2110.12038 https://arxiv.org/abs/2110.12038

[32] Udit Paul, Jiamo Liu, David Farias-llerenas, Vivek Adarsh, Arpit Gupta, and Elizabeth Belding. 2022. Characterizing Internet Access and Quality Inequities in California M-Lab Measurements. In *Proceedings of the 5th ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS '22).* Association for Computing Machinery, New York, NY, USA, 257–265. https://doi.org/10.1145/3530190.3534813

[33] Udit Paul, Jiamo Liu, Mengyang Gu, Arpit Gupta, and Elizabeth Belding. 2022. The importance of contextualization of crowdsourced active speed test measurements. In *Proceedings of the 22nd ACM Internet Measurement Conference (IMC '22).* Association for Computing Machinery, New York, NY, USA, 274–289. https://doi.org/10.1145/3517745.3561441

[34] Lin Quan, John Heidemann, and Yuri Pradkin. 2014. When the internet sleeps: correlating diurnal networks with external factors. In *Proceedings of the 2014 Conference on Internet Measurement Conference (IMC '14)*. Association for Computing Machinery, New York, NY, USA, 87–100. https://doi.org/10.1145/2663716.2663721

[35] Stephen W. Raudenbush. 1989. The analysis of longitudinal, multilevel data. *International Journal of Educational Research* 13, 7 (1989), 721–740. https://doi.org/10.1016/0883-0355(89)90024-4

[36] Stephen W. Raudenbush and Anthony S. Bryk. 2002. *Hierarchical Linear Models: Applications and Data Analysis Methods*. SAGE Publications. https://books.google.com/books?id=uyCV0CNGDLQC

[37] Robert J. Sampson, Stephen W. Raudenbush, and Felton Earls. 1997. Neighborhoods and Violent Crime: A Multilevel Study of Collective Efficacy. *Science* 277, 5328 (1997), 918–924. https://doi.org/10.1126/science.277.5328.918 arXiv:https://www.science.org/doi/pdf/10.1126/science.277.5328.918

[38] Robert J. Sampson, Patrick Sharkey, and Stephen W. Raudenbush. 2008. Durable effects of concentrated disadvantage on verbal ability among African-American children. *Proceedings of the National Academy of Sciences* 105, 3 (2008), 845–852. https://doi.org/10.1073/pnas.0710189104 arXiv:https://www.pnas.org/doi/pdf/10.1073/pnas.0710189104

[39] Reinaldo Sanchez-Arias, Luis G. Jaimes, Shahram Taj, and Md. Selim Habib. 2023. Understanding the State of Broadband Connectivity: An Analysis of Speedtests and Emerging Technologies. *IEEE Access* 11 (2023), 101580–101603. https://doi.org/10.1109/ACCESS.2023.3313231

[40] James Saxon and Dan A. Black. 2022. What we can learn from selected, unmatched data: Measuring internet inequality in Chicago. *Computers, Environment and Urban Systems* 98 (2022), 101874. https://doi.org/10.1016/j.compenvurbsys.2022.101874

[41] Ranya Sharma, Tarun Mangla, James Saxon, Marc Richardson, Nick Feamster, and Nicole P Marwell. 2022. Benchmarks or Equity? A New Approach to Measuring Internet Performance. *A New Approach to Measuring Internet Performance (August 3, 2022)* (2022).

[42] Taveesh Sharma, Paul Schmitt, Francesco Bronzino, Nick Feamster, and Nicole P. Marwell. 2024. Beyond Data Points: Regionalizing Crowdsourced Latency Measurements. *Proc. ACM Meas. Anal. Comput. Syst.* 8, 3, Article 34 (Dec. 2024), 24 pages. https://doi.org/10.1145/3700416

[43] Marco R. Steenbergen and Bradford S. Jones. 2002. Modeling Multilevel Data Structures. *American Journal of Political Science* 46, 1 (2002), 218–237. http://www.jstor.org/stable/3088424

[44] Kaixin Sui, Mengyu Zhou, Dapeng Liu, Minghua Ma, Dan Pei, Youjian Zhao, Zimu Li, and Thomas Moscibroda. 2016. Characterizing and Improving WiFi Latency in Large-Scale Operational Networks. In *Proceedings of the 14th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys '16)*. Association for Computing Machinery, New York, NY, USA, 347–360. https://doi.org/10.1145/2906388.2906393

[45] Srikanth Sundaresan, Walter de Donato, Nick Feamster, Renata Teixeira, Sam Crawford, and Antonio Pescapè. 2011. Broadband internet performance: a view from the gateway. In *Proceedings of the ACM SIGCOMM 2011 Conference (SIGCOMM '11)*. Association for Computing Machinery, New York, NY, USA, 134–145. https://doi.org/10.1145/2018436.2018452

[46] Srikanth Sundaresan, Xiaohong Deng, Yun Feng, Danny Lee, and Amogh Dhamdhere. 2017. Challenges in inferring internet congestion using throughput measurements. In *Proceedings of the 2017 Internet Measurement Conference (IMC '17)*. Association for Computing Machinery, New York, NY, USA, 43–56. https://doi.org/10.1145/3131365.3131382

[47] Srikanth Sundaresan, Nazanin Magharei, Nick Feamster, Renata Teixeira, and Sam Crawford. 2013. Web performance bottlenecks in broadband access networks. In *Proceedings of the ACM SIGMETRICS/International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS '13)*. Association for Computing Machinery, New York, NY, USA, 383–384. https://doi.org/10.1145/2465529.2465745

[48] Uber Technologies, Inc. 2018. H3: Uber's Hexagonal Hierarchical Spatial Index. https://h3geo.org/. (2018). Accessed: 2024-12-12.

[49] Xiao Sophia Wang, Aruna Balasubramanian, Arvind Krishnamurthy, and David Wetherall. 2014. How Speedy is SPDY?. In *Proceedings of the 11th USENIX Conference on Networked Systems Design and Implementation (NSDI'14)*. USENIX Association, Berkeley, CA, USA, 387–399. https://www.usenix.org/conference/nsdi14/technical-sessions/presentation/wang_xiao_sophia

[50] Geoffrey T. Wodtke, David J. Harding, and Felix Elwert. 2011. Neighborhood Effects in Temporal Perspective: The Impact of Long-Term Exposure to Concentrated Disadvantage on High School Graduation. *American Sociological Review* 76, 5 (2011), 713–736. https://doi.org/10.1177/0003122411420816 arXiv:https://doi.org/10.1177/0003122411420816 PMID: 22879678.

[51] Geoffrey T. Wodtke, Sagi Ramaj, and Jared Schachner. 2022. Toxic Neighborhoods: The Effects of Concentrated Poverty and Environmental Lead Contamination on Early Childhood Development. *Demography* 59, 4 (08 2022), 1275–1298. https://doi.org/10.1215/00703370-10047481 arXiv:https://read.dukeupress.edu/demography/article-pdf/59/4/1275/1624632/1275wodtke.pdf